

CONCLUSION

In this paper, we studied the distributed inference for edge natural language processing, focusing on the typical next-word prediction service. We cast the key model selection problem as an online constrained optimization problem with multiple objectives. Specifically, the problem requires simultaneously maximization of prediction click-through rate, minimization of user impatience, and guarantee of within-budget energy consumption. From the lens of constrained bandit learning, we proposed DONUT to solve the model selection problem via an integration of online learning and online control. Theoretical analysis showed that DONUT achieves sub-linear regret, ensures bounded user impatience, and maintains energy consumption with budget. Simulation results verified the outperformance of DONUT over other online algorithms in terms of the proposed integrated metric by an average 160% increase.